# MSE 311 Thermodynamics of materials-Homework 12

*Name:* *Student ID:*

*Please hand in this homework before class on December 8th.*

1. (a) Deliver the following equation based on the equation of sipnodal curve, where $\Omega = \Omega^0 + \Omega^1 T$ (b) Then deliver its first and second differential with respect to $X_A$. (Note: you must give detailed steps, otherwise you cannot get scores).

$$2\left(\Omega^0+\Omega^1 T\right)X_A\left(1-X_A\right)=RT$$

*(a)*

$$-2\Omega+RT\left(\frac{1}{X_A}+\frac{1}{X_B}\right)=0 \Rightarrow RT\frac{X_A+X_B}{X_AX_B}=2\Omega$$

$$2\left(\Omega^0+\Omega^1 T\right)X_A\left(1-X_A\right)=RT$$

$$2\left(\Omega^0+\Omega^1 T\right)X_A\left(1-X_A\right)=RT$$

*(b)*

$$2\left[\Omega^1\frac{\partial T}{\partial X_A}X_A\left(1-X_A\right)+\left(\Omega^0+\Omega^1 T\right)\left(1-X_A\right)-\left(\Omega^0+\Omega^1 T\right)X_A\right]=R\frac{\partial T}{\partial X_A}$$

$$2\Omega^1 X_A\left(1-X_A\right)\frac{\partial T}{\partial X_A}-4\left(\Omega^0+\Omega^1 T\right)X_A+2\left(\Omega^0+\Omega^1 T\right)=R\frac{\partial T}{\partial X_A}$$

$$2\Omega^1 X_A\left(1-X_A\right)\frac{\partial T}{\partial X_A}-4\left(\Omega^0+\Omega^1 T\right)X_A+2\left(\Omega^0+\Omega^1 T\right)=R\frac{\partial T}{\partial X_A}$$

$$2\Omega^1\left(1-X_A\right)\frac{\partial T}{\partial X_A}-2\Omega^1 X_A\frac{\partial T}{\partial X_A}+2\Omega^1 X_A\left(1-X_A\right)\frac{\partial^2 T}{\partial X_A^2}-4\Omega^1\frac{\partial T}{\partial X_A}X_A-4\left(\Omega^0+\Omega^1 T\right)+2\Omega^1\frac{\partial T}{\partial X_A}=R\frac{\partial^2 T}{\partial X_A^2}$$

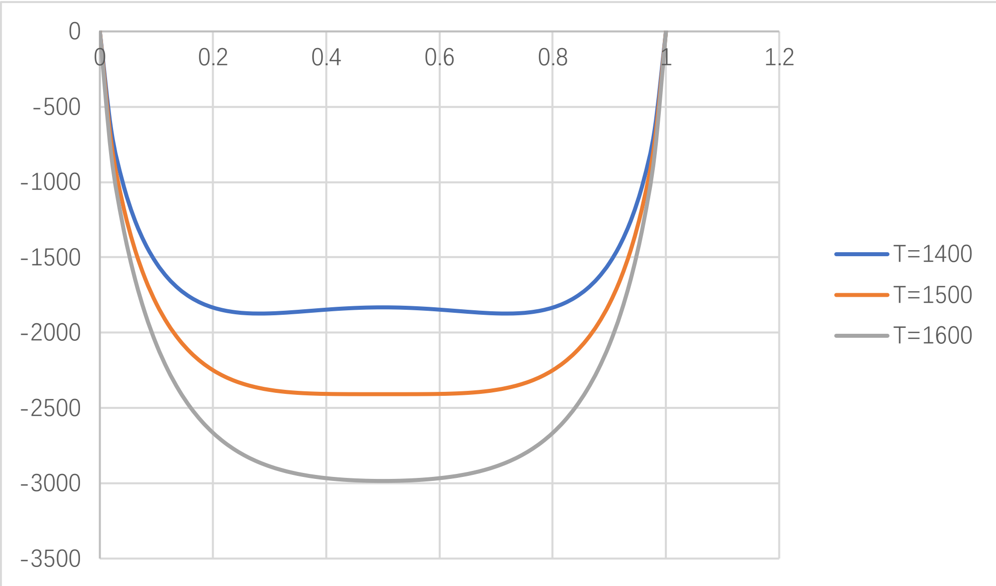
$$2\Omega^1 X_A\left(1-X_A\right)\frac{\partial^2 T}{\partial X_A^2}+4\Omega^1\frac{\partial T}{\partial X_A}-8\Omega^1 X_A\frac{\partial T}{\partial X_A}-4\left(\Omega^0+\Omega^1 T\right)=R\frac{\partial^2 T}{\partial X_A^2}$$

$$2\Omega^1 X_A\left(1-X_A\right)\frac{\partial^2 T}{\partial X_A^2}+4\Omega^1\left(1-2X_A\right)\frac{\partial T}{\partial X_A}-4\left(\Omega^0+\Omega^1 T\right)=R\frac{\partial^2 T}{\partial X_A^2}$$

2. The free energy of mixing of a regular solution is given by ($\Omega$ = 24,943 J/mole)

$$\Delta G_{mix}=\Omega X_AX_B+RT\left(X_A\ln X_A+X_B\ln X_B\right)$$

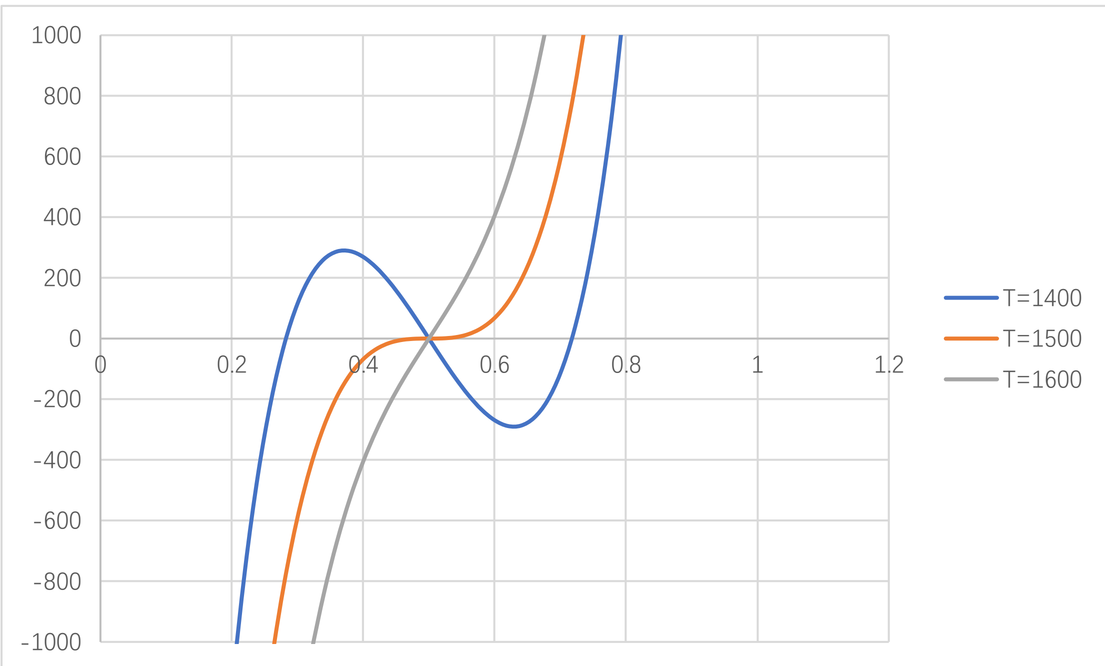
a. Plot $\Delta G_{mix}$ versus $X_B$ at 1400, 1500, and 1600 K.
b. Plot $\partial\Delta G_{mix} / \partial X_B$ versus $X_B$ at the same temperatures as (a).
c. Determine the critical temperature for this alloy. Show your work.




a)

$$\Delta G_{mix} = \Omega X_A X_B + RT\left(X_A \ln X_A + X_B \ln X_B\right)$$
$$= 24943(1-X_B)X_B + 8.314T\left[(1-X_B)\ln(1-X_B) + X_B \ln X_B\right]$$

b) $$\frac{\partial \Delta G_{mix}}{\partial X_B} = 24943(1-2X_B) + 8.314T\left[-\ln(1-X_B) - 1 + \ln X_B + 1\right]$$
$$= 24943(1-2X_B) + 8.314T\left[\ln X_B - \ln(1-X_B)\right]$$




*c) 1499 K*

3. Please draw the Gibbs free energy of mixing curves at various temperatures, and the

phase diagram for a binary system which forms regular solid solutions in which $\Omega_s$ = 10,000 J and regular liquid solutions in which $\Omega_l$ = -5,000 J. The melting temperatures of A and B are, respectively, 800 and 1200 K.

$$\Delta G^{\circ}_{m,(A)} = 8000 - 10T$$

$$\Delta G^{\circ}_{m,(B)} = 12000 - 10T$$

Ps: You should at least draw 5 different temperatures mixing curves and the final phase diagram, at least 6 pictures. You could use the software to simplify your calculate.

*When* $T_{m,A} < T < T_{m,B}$

$$\Delta G_l^M = X_B \Delta G^{\circ}_{m,(B)} + RT\left(X_A \ln X_A + X_B \ln X_B\right) + \Omega_l X_A X_B$$

$$= \left(12000 - 10T\right) X_B + 8.314T\left[\left(1 - X_B\right)\ln\left(1 - X_B\right) + X_B \ln X_B\right] - 5000\left(1 - X_B\right) X_B$$

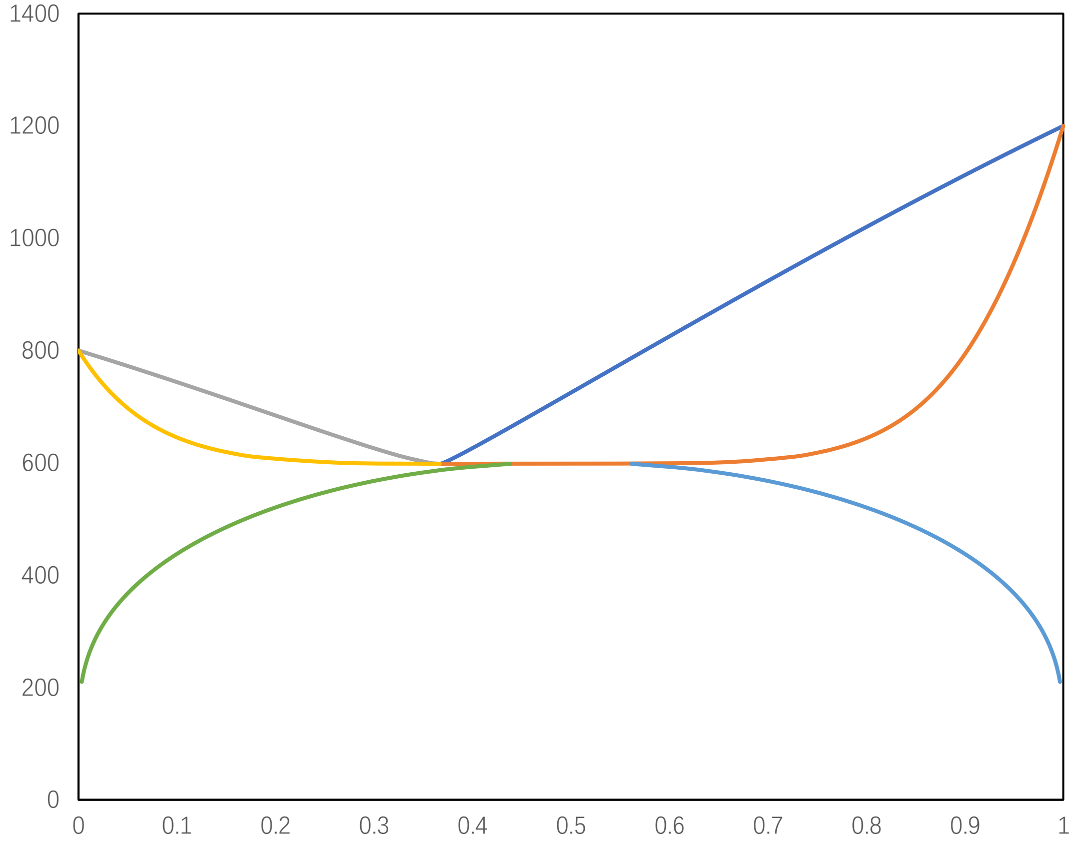
$$\frac{\partial \Delta G_l^M}{\partial X_B} = 12000 - 10T + 8.314T\left[\ln X_B - \ln\left(1 - X_B\right)\right] - 5000\left(1 - 2X_B\right) = 7000 - 10T + 8.314T\left[\ln X_B - \ln\left(1 - X_B\right)\right] + 1000$$

$$\Delta G_s^M = -X_A \Delta G^{\circ}_{m,(A)} + RT\left(X_A \ln X_A + X_B \ln X_B\right) + \Omega_s X_A X_B$$

$$= \left(10T - 8000\right)\left(1 - X_B\right) + 8.314T\left[\left(1 - X_B\right)\ln\left(1 - X_B\right) + X_B \ln X_B\right] + 10000\left(1 - X_B\right) X_B$$

$$\frac{\partial \Delta G_s^M}{\partial X_B} = 8000 - 10T + 8.314T\left[\ln X_B - \ln\left(1 - X_B\right)\right] + 10000\left(1 - 2X_B\right) = 18000 - 10T + 8.314T\left[\ln X_B - \ln\left(1 - X_B\right)\right] - 200$$

$$\begin{cases} 7000 - 10T + 8.314T\left[\ln X_{B,l} - \ln\left(1 - X_{B,l}\right)\right] + 10000 X_{B,l} = 18000 - 10T + 8.314T\left[\ln X_{B,s} - \ln\left(1 - X_{B,s}\right)\right] - 20000 X_{B,s} \\ \Delta G_l^M - \dfrac{\partial \Delta G_l^M}{\partial X_{B,l}} X_{B,l} = \Delta G_s^M - \dfrac{\partial \Delta G_s^M}{\partial X_{B,s}} X_{B,s} \end{cases}$$


1400
1200
1000
800
600
400
200
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1